

NETWORKING ASSIGNMENT REPORT

OSPF Routing Configuration

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Simulation Tool: Cisco Packet Tracer

1. Objective

This assignment extends an existing Head Office / CTG-branch network by adding a new Khulna branch and a dedicated Internet zone, then enabling OSPF so the new sites and the Head Office core exchange routes dynamically. The specific goals were:

- Build a new branch site, Khulna, with its own LAN and a WAN link back to the Head Office core.
- Build an Internet zone consisting of an internal Internet-facing router and an ISP router, giving the network a path to an external "Google" server.
- Configure OSPF on the Khulna branch router, the Internet router, the ISP router, and core-RTR1 so all four exchange routes dynamically.
- Verify end-to-end reachability from a Head Office LAN host to the external Google server.
- Verify end-to-end reachability from a Head Office LAN host to the Khulna branch.
- Capture and verify the routing table of every router in scope to confirm OSPF convergence.

2. Network Topology

The diagram below shows the completed topology as built in Cisco Packet Tracer. The Head Office core (core-RTR1) sits at the centre, with the pre-existing CTG branch and static-routed LAN to its left, the newly built Khulna branch above it, and the new Internet zone (INT-RTR → isp-rtr → Google server) to the right. Below the core sit the Head Office multilayer switches and six VLANs (10, 20, 30, 40, 50 and 60) serving the Head Office LAN hosts.

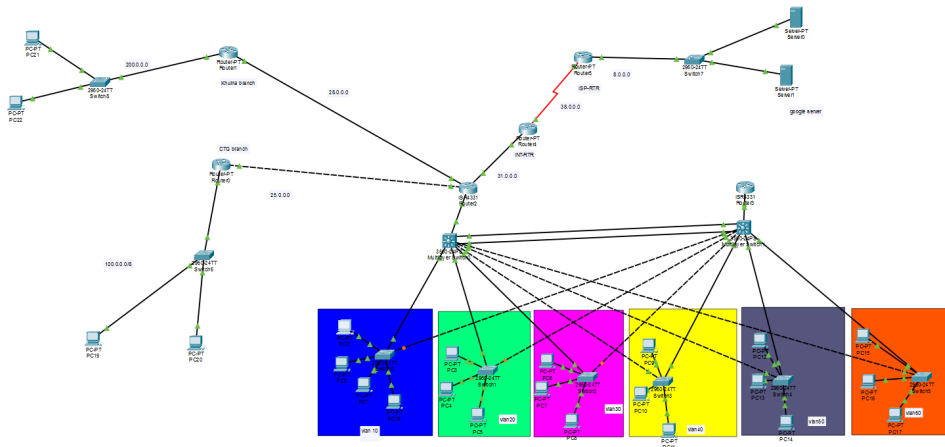


Figure 1. Complete network topology, including the Khulna branch and Internet zone.

2.1 Device Inventory

The table below summarises the routers relevant to the OSPF scope of this assignment (the CTG branch is included for context, as it peers with core-RTR1 via static routes rather than OSPF).

Device	Role	Key Interfaces / Networks
CTG-RTR (Router0)	CTG Branch router (static routing)	LAN 100.0.0.0/8; WAN 25.0.0.0/8 to core-RTR1
khulna-RTR (Router1)	Khulna Branch router (OSPF)	LAN 200.0.0.0/8; WAN 28.0.0.0/8 to core-RTR1
core-RTR1 (Router2)	Head Office core router / OSPF backbone	VLAN 10/20/30 sub-interfaces, 25.0.0.0/8 to CTG, 28.0.0.0/8 to Khulna, 31.0.0.0/8 to INT-RTR
INT-RTR (Router4)	Internet edge router (OSPF)	31.0.0.0/8 to core-RTR1; 38.0.0.0/8 to isp-rtr
isp-rtr (Router5)	ISP router (OSPF)	38.0.0.0/8 to INT-RTR; 8.0.0.0/8 to Google server LAN

3. Implementation

3.1 Building the Khulna Branch

A new branch router (khulna-RTR) was added with its own access-layer switch and two end hosts on the 200.0.0.0/8 LAN. A point-to-point link on the 28.0.0.0/8 network connects khulna-RTR to core-RTR1, giving the branch a single path into the Head Office core.

3.2 Building the Internet Zone

An Internet zone was created using two routers: INT-RTR, which faces the internal network, and isp-rtr, which represents the service provider edge. INT-RTR connects to core-RTR1 over the 31.0.0.0/8 network and to isp-rtr over a 38.0.0.0/8 serial link. isp-rtr in turn connects to a switch and two servers on the 8.0.0.0/8 network, one of which represents the external Google server used for reachability testing.

3.3 OSPF Configuration

OSPF was enabled on khulna-RTR, INT-RTR, isp-rtr and core-RTR1 so that the Khulna branch and the Internet zone are learned dynamically rather than through static routes. This is verified indirectly and conclusively by the routing tables in Section 4: each of these four routers shows the other routers' networks tagged with the OSPF code "O", and the reported metrics and next-hops are consistent with the physical topology (e.g. core-RTR1 shows a nearly identical cost to 8.0.0.0/8 whether reached via its own OSPF process or cross-checked against INT-RTR and isp-rtr's tables). The CTG branch was left on static routing on both core-RTR1 and CTG-RTR, since it was outside the OSPF scope defined for this assignment.

4. Verification and Testing

4.1 Routing Table Verification

The routing table of every router in scope was captured with "sh ip route" to confirm OSPF convergence. In each table, routes marked "O" are learned dynamically via OSPF and routes marked "C" / "L" are directly

connected — the presence of “O” entries for every remote network confirms OSPF is running and fully converged.

khulna-RTR — learns all Head Office and Internet-zone networks via OSPF over its 28.0.0.0/8 link to core-RTR1:

```
khulna-RTR#
khulna-RTR#
khulna-RTR#
khulna-RTR#
khulna-RTR#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

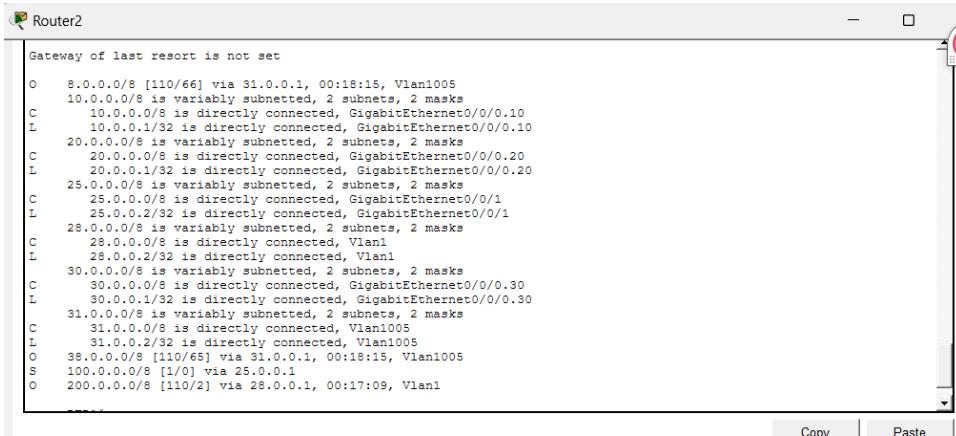
Gateway of last resort is not set

O    8.0.0.0/8 [110/67] via 28.0.0.2, 00:19:49, FastEthernet1/0
O    10.0.0.0/8 [110/2] via 28.0.0.2, 00:19:49, FastEthernet1/0
O    20.0.0.0/8 [110/2] via 28.0.0.2, 00:19:49, FastEthernet1/0
O    25.0.0.0/8 [110/2] via 28.0.0.2, 00:19:22, FastEthernet1/0
C    28.0.0.0/8 is directly connected, FastEthernet1/0
O    30.0.0.0/8 [110/2] via 28.0.0.2, 00:19:49, FastEthernet1/0
O    31.0.0.0/8 [110/2] via 28.0.0.2, 00:19:49, FastEthernet1/0
O    38.0.0.0/8 [110/66] via 28.0.0.2, 00:19:49, FastEthernet1/0
O    100.0.0.0/8 [110/3] via 28.0.0.2, 00:19:22, FastEthernet1/0
C    200.0.0.0/8 is directly connected, FastEthernet0/0

khulna-RTR#
```

Figure 2. khulna-RTR routing table.

core-RTR1 — the Head Office core shows OSPF routes to the Khulna LAN (200.0.0.0/8), the Internet zone (8.0.0.0/8, 38.0.0.0/8) and its own directly connected VLAN and WAN interfaces; the CTG LAN (100.0.0.0/8) appears as a static route, as expected:

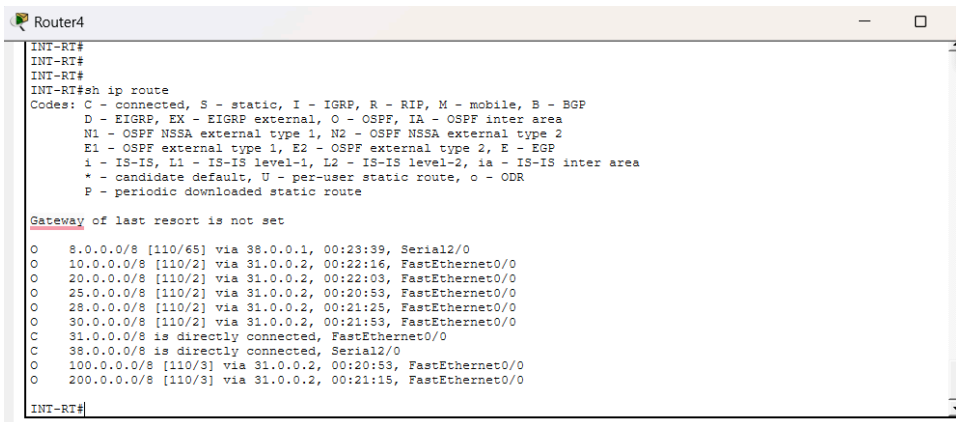


```
Router2
Gateway of last resort is not set

O    8.0.0.0/8 [110/66] via 31.0.0.1, 00:18:15, Vlan1005
C    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    10.0.0.0/8 is directly connected, GigabitEthernet0/0/0.10
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0/0.10
C    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    20.0.0.0/8 is directly connected, GigabitEthernet0/0/0.20
L    20.0.0.1/32 is directly connected, GigabitEthernet0/0/0.20
C    25.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    25.0.0.0/8 is directly connected, GigabitEthernet0/0/1
L    25.0.0.2/32 is directly connected, GigabitEthernet0/0/1
C    28.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    28.0.0.0/8 is directly connected, Vlan1
L    28.0.0.2/32 is directly connected, Vlan1
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    30.0.0.0/8 is directly connected, GigabitEthernet0/0/0.30
L    30.0.0.1/32 is directly connected, GigabitEthernet0/0/0.30
C    31.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    31.0.0.0/8 is directly connected, Vlan1005
L    31.0.0.2/32 is directly connected, Vlan1005
O    38.0.0.0/8 [110/65] via 31.0.0.1, 00:18:15, Vlan1005
S    100.0.0.0/8 [1/0] via 25.0.0.1
O    200.0.0.0/8 [110/2] via 28.0.0.1, 00:17:09, Vlan1
```

Figure 3. core-RTR1 complete routing table.

INT-RTR — confirms OSPF adjacency with both core-RTR1 and isp-rtr, with full visibility of every network in the topology:



```

Router4
INT-RTR#
INT-RTR#
INT-RTR#
INT-RTR#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

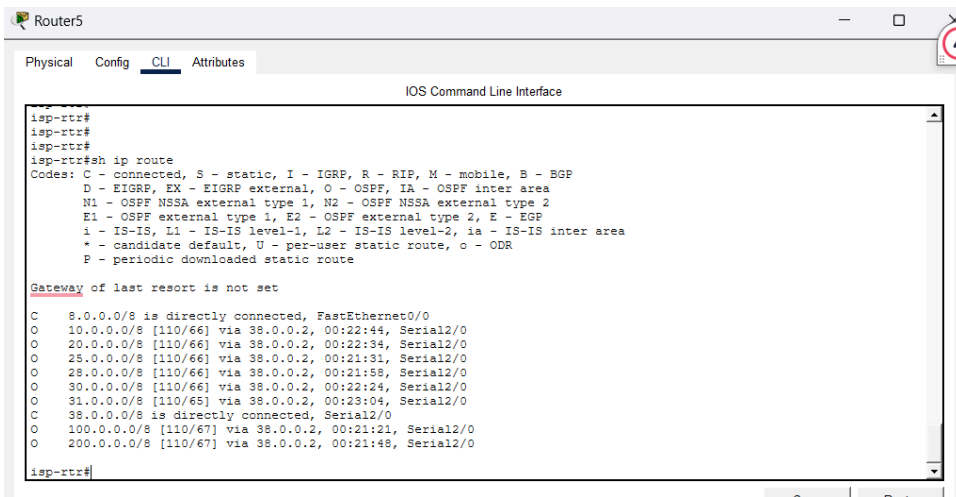
O    8.0.0.0/8 [110/65] via 38.0.0.1, 00:23:39, Serial2/0
O   10.0.0.0/8 [110/2] via 31.0.0.2, 00:22:16, FastEthernet0/0
O   20.0.0.0/8 [110/2] via 31.0.0.2, 00:22:03, FastEthernet0/0
O   25.0.0.0/8 [110/2] via 31.0.0.2, 00:20:53, FastEthernet0/0
O   28.0.0.0/8 [110/2] via 31.0.0.2, 00:21:25, FastEthernet0/0
O   30.0.0.0/8 [110/2] via 31.0.0.2, 00:21:59, FastEthernet0/0
C   31.0.0.0/8 is directly connected, FastEthernet0/0
C   38.0.0.0/8 is directly connected, Serial2/0
O  100.0.0.0/8 [110/3] via 31.0.0.2, 00:20:53, FastEthernet0/0
O  200.0.0.0/8 [110/3] via 31.0.0.2, 00:21:15, FastEthernet0/0

INT-RTR#

```

Figure 4. INT-RTR routing table.

isp-rtr — at the network edge, learns every internal network (including the Khulna and CTG LANs) via OSPF through INT-RTR:



```

Router5
Physical Config CLI Attributes
IOS Command Line Interface

isp-rtr#
isp-rtr#
isp-rtr#
isp-rtr#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

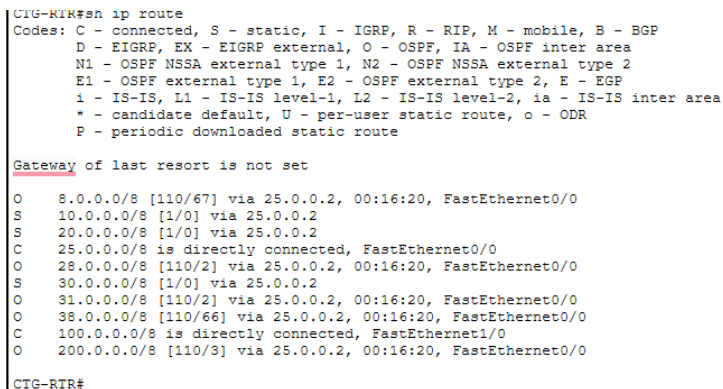
C    8.0.0.0/8 is directly connected, FastEthernet0/0
O   10.0.0.0/8 [110/66] via 38.0.0.2, 00:22:44, Serial2/0
O   20.0.0.0/8 [110/66] via 38.0.0.2, 00:22:34, Serial2/0
O   25.0.0.0/8 [110/66] via 38.0.0.2, 00:21:31, Serial2/0
O   28.0.0.0/8 [110/66] via 38.0.0.2, 00:21:58, Serial2/0
O   30.0.0.0/8 [110/66] via 38.0.0.2, 00:22:24, Serial2/0
O   31.0.0.0/8 [110/65] via 38.0.0.2, 00:23:04, Serial2/0
C   38.0.0.0/8 is directly connected, Serial2/0
O  100.0.0.0/8 [110/67] via 38.0.0.2, 00:21:21, Serial2/0
O  200.0.0.0/8 [110/67] via 38.0.0.2, 00:21:48, Serial2/0

isp-rtr#

```

Figure 5. isp-rtr routing table.

CTG-RTR — shown for completeness; this branch continues to use static routes rather than OSPF, so its table lists “S” entries alongside its OSPF-learned 8.0.0.0/8 and 38.0.0.0/8 networks (learned because core-RTR1 redistributes/advertises them into the link it shares with CTG-RTR):



```

CTG-RTR#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

O    8.0.0.0/8 [110/67] via 25.0.0.2, 00:16:20, FastEthernet0/0
S   10.0.0.0/8 [1/0] via 25.0.0.2
S   20.0.0.0/8 [1/0] via 25.0.0.2
C   25.0.0.0/8 is directly connected, FastEthernet0/0
O   28.0.0.0/8 [110/2] via 25.0.0.2, 00:16:20, FastEthernet0/0
S   30.0.0.0/8 [1/0] via 25.0.0.2
O   31.0.0.0/8 [110/2] via 25.0.0.2, 00:16:20, FastEthernet0/0
O   38.0.0.0/8 [110/66] via 25.0.0.2, 00:16:20, FastEthernet0/0
C  100.0.0.0/8 is directly connected, FastEthernet1/0
O  200.0.0.0/8 [110/3] via 25.0.0.2, 00:16:20, FastEthernet0/0

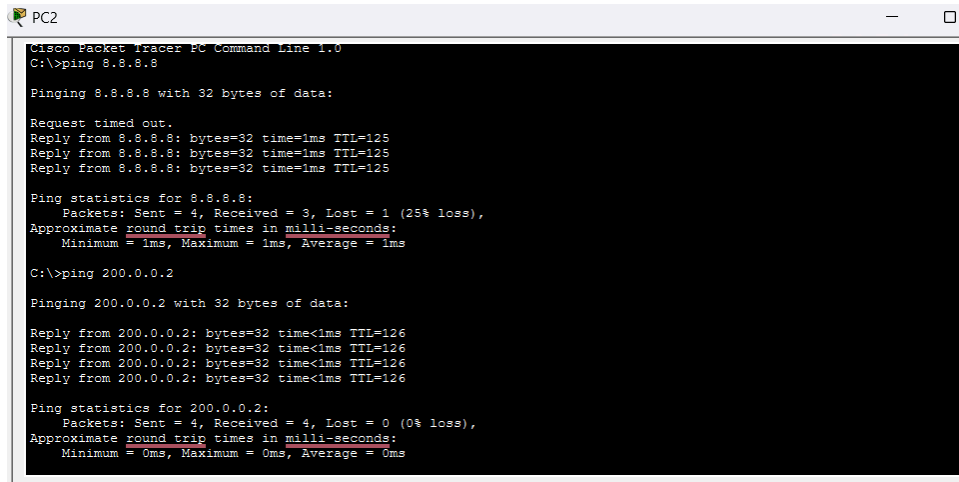
CTG-RTR#

```

Figure 6. CTG-RTR routing table.

4.2 Reachability Testing

From PC2 on the Head Office LAN, ping tests were run against the external Google server (8.8.8.8) and against a host on the Khulna branch LAN (200.0.0.2). Both tests succeeded, confirming end-to-end reachability in both directions across the OSPF-routed backbone.



```

PC2
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Reply from 200.0.0.2: bytes=32 time<1ms TTL=126
Reply from 200.0.0.2: bytes=32 time<1ms TTL=126
Reply from 200.0.0.2: bytes=32 time<1ms TTL=126
Reply from 200.0.0.2: bytes=32 time<1ms TTL=126

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
  
```

Figure 7. PC2 ping results to the Google server (8.8.8.8) and to the Khulna branch (200.0.0.2).

The ping to 8.8.8.8 shows one initial timeout followed by three successful replies (75% success), a normal ARP-resolution delay on the first packet rather than a routing fault. The ping to 200.0.0.2 returned four out of four replies with 0% loss, confirming a fully converged path to the Khulna branch.

5. Results Summary

#	Task	Status	Evidence
1	Build Khulna branch	Complete	Figure 1; Figure 2 (khulna-RTR routing table)
2	Build Internet zone (INT-RTR, isp-rtr)	Complete	Figure 1; Figures 4–5
3	Configure OSPF on Khulna, INT-RTR, isp-rtr, core-RTR1	Complete	Figures 2–5 (OSPF “O” routes present on all four)
4	Verify Head Office LAN → Google server reachability	Complete	Figure 7 (ping to 8.8.8.8)
5	Verify Head Office LAN → Khulna branch reachability	Complete	Figure 7 (ping to 200.0.0.2)
6	Verify routing table of each router	Complete	Figures 2, 3, 4, 5, 6

6. Conclusion

The Khulna branch and Internet zone were successfully added to the existing network, and OSPF was enabled across khulna-RTR, INT-RTR, isp-rtr and core-RTR1. Captured routing tables confirm every router in scope has full, dynamically learned visibility of the rest of the network, and ping testing confirms working

end-to-end connectivity from the Head Office LAN to both the Khulna branch and the external Google server. All six assignment objectives were met.